

ProcureSight: An Auditable Procurement Spend Analytics and Opportunity Prioritization System

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Abstract— ProcureSight is a Phase 1 local procurement decision-support system that transforms validated purchase-order data into spend analytics, internal price benchmarks, deterministic review opportunities, and optional grounded advisory narratives. The implemented workflow accepts a defined CSV structure through command-line ingestion, validates and normalizes the records into PostgreSQL, computes SQL-first analytics, derives supplier-level weighted prices and item-level continuous P25 benchmarks, and detects deterministic price optimization, contract leakage, supplier consolidation, and supplier performance opportunities. The user interface is implemented in React and TypeScript, while the backend uses FastAPI, SQLAlchemy, Alembic, and PostgreSQL running locally through Docker Compose. The advisory layer uses the OpenAI Responses API with a configurable gpt-5.4-mini default and deterministic fallback; financial values are computed in the backend before any narrative is generated. On the supplied 47,128-record dataset, the workflow processed 106 suppliers, 10 countries, 10 categories, 50 items, and 5 business units with total spend of USD 411,183,335.24. The isolated full-engine execution produced 50 internal benchmarks and 254 active opportunities. The USD 12,512,042.6726 result is strictly a price-optimization estimated potential derived from internal comparison rules; it is not a realized saving or an external-market benchmark. Dataset provenance is not verified from the repository, so the dataset is not attributed to a company or external source. The implementation demonstrates reproducible candidate generation and explainable decision support while retaining human review for action and savings realization.

Keywords— *procurement analytics; spend analysis; price benchmarking; cost optimization; supplier performance; contract compliance; deterministic analytics; decision support; FastAPI; PostgreSQL; React; grounded LLM.*

I. INTRODUCTION

Procurement purchase-order data contains information about suppliers, items, quantities, prices, contract status, quality, delivery, and operating units. Converting these transactions into actionable procurement insight requires more than reporting aggregate spend: users must be able to identify material cost or compliance issues, compare prices using a consistent internal reference, prioritize findings, and review the evidence behind each candidate.

ProcureSight implements a traceable workflow from structured purchase-order data to reviewable procurement

candidates. A core design principle is that deterministic services calculate financial and operational context first; the optional language-model layer is restricted to explaining verified context. Consequently, the system does not assign financial calculation, validation, or final procurement decisions to generative output.

The Phase 1 scope is intentionally local and bounded. Data is ingested through a command-line CSV workflow, stored in PostgreSQL, analyzed using SQL/Python services, and exposed through FastAPI to a React/Vite frontend. External market benchmarks, foreign-exchange conversion, forecasting, anomaly detection, scheduled ingestion, authentication/authorization, realized-savings workflows, and production operations are outside the implemented scope.

II. RELATED WORK AND RESEARCH GAP

Recent research on procurement and supply-chain decision support emphasizes supplier evaluation, multi-criteria decision-making, transparency, risk assessment, consistent criteria weighting, and optimization. The following studies were selected in the project review because they provide complementary perspectives on these themes.

A. Haffar and Özceylan—Supplier Selection

Haffar and Özceylan [1] proposed a blockchain-based system for supplier selection in sustainable and agile supply chains. Their system combines multiple supplier-selection criteria with a multi-attribute reverse auction and uses blockchain to improve transparency and traceability. The study demonstrates how technology-supported supplier evaluation can preserve decision evidence. ProcureSight adopts the principle of traceable evidence but uses normalized purchase-order data and deterministic analytical calculations rather than blockchain or reverse auction mechanisms.

B. Runtuk et al.—Multi-Criteria Supplier Ranking

Runtuk et al. [2] compared Fuzzy VIKOR and Fuzzy MOORA for performance-based supplier selection. Their work highlights the role of multiple criteria, criteria weighting, sensitivity, and transparent supplier ranking. ProcureSight similarly evaluates supplier performance using explicit price, quality, delivery, and contract components, but uses deterministic configurable weights

rather than the specific fuzzy MCDM techniques studied in that work.

C. Qiu et al.—Supply-Chain Risk Assessment

Qiu et al. [3] developed a strategic decision-support approach for supply-chain risk management using probabilistic linguistic term sets and an extended CRADIS method. The study addresses uncertain or subjective judgments through structured multi-criteria risk prioritization. ProcureSight does not implement probabilistic or predictive risk modeling in Phase 1; instead, it uses verified operational indicators such as quality rejection, late delivery, and contract compliance.

D. Moretti et al.—Sustainable Supplier Selection

Moretti et al. [4] introduced the Least-to-Most Range (LMR) method for consistent weighting in green supplier selection. The work emphasizes explicit representation of criteria importance and consistency when multiple criteria are considered. ProcureSight uses configurable deterministic weights and thresholds to maintain an interpretable and reproducible scoring process, rather than implementing the LMR method itself.

E. Shahsavary et al.—Machine-Learning-Based Supply-Chain Risk

Shahsavary et al. [5] investigated risk prediction in distributed supply-chain design using machine learning and bi-objective optimization. Their approach combines Random Forest-based risk prediction with optimization to balance cost and risk. The study motivates possible future predictive extensions for ProcureSight, but Phase 1 deliberately remains focused on verified historical procurement metrics and deterministic opportunity detection.

F. Research Gap and Project Contribution

The reviewed studies primarily address specific decision-support objectives such as supplier selection, risk prioritization, sustainable supplier evaluation, or predictive optimization. A practical procurement workflow also needs these analytical activities to be connected to transactional procurement data and operational spend information. This motivates an integrated workflow that can transform purchase-order data into measurable procurement insights, identify potential cost and compliance issues, prioritize findings, and provide interpretable recommendations.

Another common requirement is traceability: rankings, risk indicators, and optimization outputs should be supported by explicit criteria and evidence so that procurement personnel can review and validate them before action. ProcureSight addresses this implementation-level gap by integrating validated purchase-order data, normalized relational storage,

deterministic spend analytics, internal price benchmarking, rule-based opportunity detection, priority scoring, and an optional grounded AI advisory layer in one workflow.

The contribution is therefore integration and traceability rather than a novel AI/ML algorithm. ProcureSight does not directly implement blockchain, Fuzzy VIKOR, Fuzzy MOORA, CRADIS, LMR weighting, or Random Forest risk prediction. These methods are treated as literature-informed design considerations. Final procurement decisions remain under human review.

Table 1. Comparison of reviewed literature and ProcureSight

Aspect	Reviewed literature	ProcureSight
Supplier evaluation	Multi-criteria supplier selection and ranking	Deterministic supplier-performance scoring
Risk assessment	Risk prioritization and predictive risk models	Verified quality, delivery and contract indicators
Criteria weighting	Fuzzy and multi-criteria weighting methods	Configurable deterministic weights
Price analysis	Supplier selection and optimization contexts	Internal supplier-weighted P25 benchmarking
Opportunity detection	Usually specific decision objectives	Four deterministic opportunity types
Prioritization	Ranking/optimization methods	Impact, confidence and severity score
AI explanation	Not the primary focus	Optional grounded LLM advisory
Traceability	Depends on individual method	Supporting metrics and context retained
Final decision	Decision-support oriented	Human review required

III. PROPOSED SYSTEM

ProcureSight follows five principal stages: (1) command-line ingestion of a defined CSV; (2) Pydantic validation and normalized PostgreSQL persistence; (3) SQL-first KPI and supplier-price analysis; (4) deterministic benchmark and opportunity generation; and (5) FastAPI delivery to a React/Vite frontend with an optional advisory layer. PostgreSQL 16 is orchestrated locally with Docker Compose.

analytics and deterministic findings. FastAPI exposes live endpoints to the React/Vite frontend. The optional LLM is invoked only for a selected opportunity context and can be replaced by a deterministic fallback.

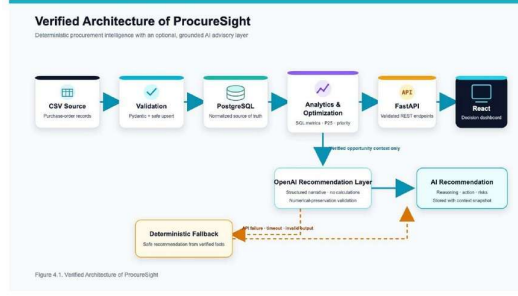


Fig. 1. Verified architecture of ProcureSight.

The architecture is batch-oriented rather than real-time. The importer validates and normalizes dimension data and upserts purchase orders. PostgreSQL stores suppliers, categories, items, business units, purchase orders, benchmarks, opportunities, and recommendations. SQL-first services and the optimization engine calculate analytics and deterministic findings. FastAPI exposes the current stored analytics to the frontend. The optional LLM receives only selected opportunity context and can be replaced by a deterministic fallback.

IV. METHODOLOGY

A. Data Handling and Analytics

The supplied dataset contains 47,128 records and 16 CSV columns: `po_id`, `order_date`, `promised_delivery_date`, `actual_delivery_date`, `supplier_id`, `supplier_name`, `supplier_country`, `category`, `item`, `business_unit`, `unit_price`, `quantity`, `line_total`, `payment_terms`, `on_contract`, and `quality_rejected`. Date fields and Decimal amounts are coerced and Boolean fields use strict accepted representations. Invalid input produces a rejection report and prevents a write. The verified execution accepted all 47,128 records with zero rejected records.

The normalized model separates suppliers, categories, items, and business units into dimensions and stores purchase orders as facts. Analytics support inclusive filtering by date, supplier, category, business unit, and country. Implemented measures include total spend, order count, average order value, spend share, concentration, monthly spend and growth, contract compliance, quality-rejected spend/rate, and late-delivery rate/delay.

B. Internal Price Benchmarking

For each item-supplier group, ProcureSight calculates a quantity-weighted supplier price. For each item, the benchmark is the continuous P25 of the resulting supplier-level weighted prices. The benchmark is an internal comparison only; it is not an external market price, does not use FX conversion, and does not apply a transaction-level cheapest-price rule.

$$WeightedPrice(s,i) = \frac{\sum[UnitPrice \times Quantity]}{\sum[Quantity]} \quad (1)$$

$$Benchmark(i) = ContinuousP25(\{WeightedPrice(s,i) \text{ for suppliers } s \text{ of item } i\}) \quad (2)$$

For a sorted set of n supplier prices, the percentile position is $(n - 1) \times \text{percentile}$, with linear interpolation between neighboring values. The benchmark stores P25, minimum, median, P75, maximum, supplier coverage, quantity, and spend.

C. Confidence and Supplier Performance

The implemented confidence score combines supplier coverage, transaction count, quantity, and price stability. The supplier-performance score combines price, quality, delivery, and contract components. The report's verified formula catalogue defines these calculations as follows.

$$Confidence = 0.35 \times SupplierFactor + 0.25 \times TransactionFactor + 0.20 \times QuantityFactor + 0.20 \times StabilityFactor \quad (3)$$

$$SupplierScore = 0.40 \times PriceScore + 0.25 \times QualityScore + 0.20 \times DeliveryScore + 0.15 \times ContractScore \quad (4)$$

$SupplierFactor = \min(\text{supplier count} / 5, 1)$; $TransactionFactor = \min(\text{transaction count} / 20, 1)$; $QuantityFactor = \min(\log_{10}(\text{quantity} + 1) / 4, 1)$; and $StabilityFactor = \max(0, 1 - \min(|P75 - P25| / \text{median}, 1))$. The underlying quality, delivery, and contract score definitions are based on the validated operational rates described in the project report.

D. Opportunity Detection and Priority Ranking

Table 2. Deterministic opportunity rules.

Opportunity type	Verified rule
Price optimization	At least 3 suppliers; variance $\geq 10\%$; estimated potential $\geq$ USD 5,000
Contract leakage	Item-supplier off-contract spend $\geq$ USD 100,000
Supplier consolidation	At least 3 suppliers; item spend $\geq$ USD 250,000; P25–P75 dispersion $\geq 15\%$
Supplier performance	Supplier spend $\geq$ USD 250,000 and performance score < 75

Opportunity types can overlap. Price-only potential is not added to contract-leakage, consolidation, or supplier-performance review spend. Priority combines financial impact, confidence, and severity. The impact factor is capped at one million US dollars for scoring.

$$Priority = 100 \times (0.50 \times ImpactFactor + 0.30 \times Confidence + 0.20 \times Severity) \quad (5)$$

Priority bands are Critical for scores ≥ 80 , High for ≥ 60 , Medium for ≥ 40 , and Low otherwise. $ImpactFactor = \min(\text{financial impact} / 1,000,000, 1)$.

E. Grounded AI Advisory Layer

The optional advisory layer uses the OpenAI Responses API with a configurable gpt-5.4-mini default. It receives selected opportunity context such as opportunity type, item, supplier, actual and benchmark prices, variance, quantity, potential or review-spend amount, confidence, supplier metrics, and supporting operational metrics. The layer does not receive the full CSV and does not calculate financial values.

Guardrails include a strong system prompt, JSON Schema validation, Pydantic validation, exact-impact equality checking, and rejection of prose numbers that are absent from the verified context. Two attempts are allowed; timeout, rate-limit, HTTP, malformed JSON, schema, or numeric-preservation failures result in a deterministic fallback recommendation. The design therefore treats the LLM as a constrained explanation layer rather than an autonomous procurement engine.

V. IMPLEMENTATION

Table 3. Implemented technology stack.

Layer	Verified technology
Backend	Python 3.11+, FastAPI, Pydantic 2, SQLAlchemy 2, Alembic
Database	PostgreSQL 16 via Docker Compose; psycopg 3
Frontend	React 19, TypeScript, Vite 7, Tailwind CSS 4, Recharts
Testing	Pytest; Vitest and Testing Library
AI	OpenAI Responses API through HTTPX with deterministic fallback

The frontend provides Dashboard, Spend Analysis, Cost Optimization, and AI Advisor pages with live API calls. The backend exposes health, dashboard, analytics, catalog, benchmark, opportunity, and recommendation resources. CSV ingestion remains CLI-only; no file-upload API is implemented. The verified deployment state is local: Docker Compose provisions PostgreSQL and Uvicorn/Vite are used locally. Production deployment, cloud hosting, CI, Kubernetes, public URLs, and production monitoring were not verified.

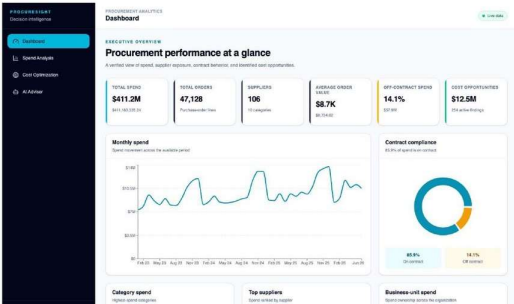


Fig. 2. Live ProcureSight dashboard from the verified project implementation.

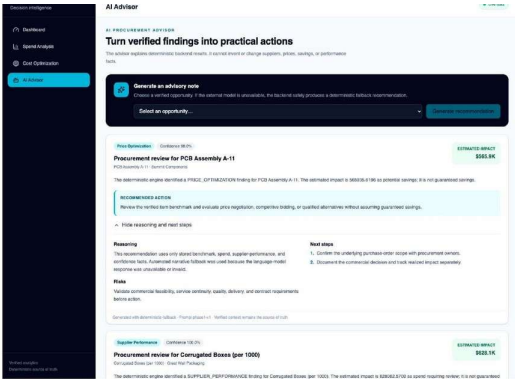


Fig. 3. AI Advisor interface using verified opportunity context.

VI. RESULTS AND DISCUSSION

Table 4. Verified execution results.

Measure	Verified result
Validated purchase orders	47,128 accepted; 0 rejected
Suppliers / countries	106 / 10
Categories / items / business units	10 / 50 / 5
Total spend	USD 411,183,335.24
Internal item benchmarks	50
Active deterministic opportunities	254
Backend tests	51 passed
Frontend tests	6 passed

Table 5. Verified opportunity results.

Opportunity measure	Verified value
Price optimization findings	107
Contract leakage findings	144
Supplier consolidation findings	1
Supplier performance findings	2
Price-only estimated potential	USD 12,512,042.6726
Price-optimization review spend	USD 108,869,726.57
Contract-leakage review spend	USD 50,428,294.10

The isolated full-engine execution confirms that the implemented pipeline can deterministically validate the supplied data, store normalized records, derive internal benchmarks, and create prioritized findings. The 254 findings do not establish business value by themselves. They are review candidates, may overlap by type, and require human investigation, commercial validation, and negotiation before any saving can be realized.

The USD 12,512,042.6726 price-only potential is derived from internal P25 comparison and configured opportunity rules. It must not be interpreted as a market-price gap, booked financial benefit, or realized saving. Similarly, review-spend values represent the spend associated with findings and are not savings.

VII. LIMITATIONS AND FUTURE WORK

Dataset provenance and authenticity are not independently verified from the project repository. The

implementation is single-currency, has no external benchmark source, and has no explicit outlier-removal procedure. Authentication, authorization, RBAC, rate limiting, production observability, backups, and production secret management are outside the current scope. No live OpenAI call was verified in the audit, and no coverage percentage or performance/load evaluation was verified.

Future work includes governed dataset provenance and authorization, role-based access control, scheduled refresh, opportunity review and realized-savings tracking, currency normalization, governed external benchmarks, production deployment and monitoring, forecasting, and anomaly detection. These extensions should be evaluated independently rather than assumed from the current Phase 1 results.

VIII. CONCLUSION

ProcureSight demonstrates a local, auditable procurement decision-support workflow that connects validated purchase-order data with deterministic analytics, internal P25 benchmarks, rule-based candidate generation, priority scoring, and optional grounded narratives. The verified implementation processed 47,128 purchase orders, produced 50 internal benchmarks, and generated 254 active findings across price optimization, contract leakage, supplier consolidation, and supplier performance.

The strongest supported contribution is traceability: procurement findings can be linked to calculated operational and financial context before human review. The project does not support claims of realized savings, external market-price validation, autonomous procurement, predictive risk capability, or production readiness.

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